Recitation : 5

An email service provider wants to classify emails as Spam or Not Spam using a linear SVM classifier.

Spam +1 , Not Spam -1

Suspicious words	links	Length	Attachment	Class
6	3	100	1	+1
1	0	350	0	-1
4	2	150	1	+1
0	0	420	0	-1
5	2	120	1	+1

Write SVM model Iteration 1

$$f(x) = w_1x_1 + x_2w_2 + w_3x_3 + w_4x_4 + b$$

Initially $w_1 \cdots w_4$ & $b = 0$, $y = +1$

$$y \cdot f(x) = 0 < 1$$

$x_1 = 6$

$x_2 = 3$

$x_4 = 100$

$x_5 = 1$

$\therefore$ Update weights & bias

$$\cancel{w = w + yx}$$

$$w = w - \eta(w - yx)$$

$$b = b + \eta y$$

$yx = (+1)(6, 3, 100, 1) = (6, 3, 100, 1)$

$w - yx = (0, 0, 0, 0) - (6, 3, 100, 1)$

$= (-6, -3, -100, -1)$

$\eta(w - yx) = (-6, -3, -100, -1)$

$w - \eta(w - yx) = (6, 3, 100, 1)$

$b = 0 + 1(1) = 1$

$\therefore w_{new} = (6, 3, 100, 1) \quad b_{new} = 1$

Iteration 2

$x = (1, 0, 350, 0) \quad y = (-1)$

$f(x) = (6 \times 1) + (3 \times 0) + (100 \times 350) \times (1 \times 0) + 1$

$= 6 + 0 + 35000 + 0 + 1 = 35007$

$y \cdot f(x) = -1(35007) = (-35007) < 1$

$yx = (-1)(1, 0, 350, 0) = (-1, 0, -350, 0)$

$w - yx = (6, 3, 100, 1) - (-1, 0, -350, 0)$

$= (7, 3, 450, 1)$

$b = 1 + 1(-1) = 0$

$\eta(w - yx)$

$= (7, 3, 450, 1)$

$w - \eta(w - yx)$

$= (6, 3, 100, 1) - (7, 3, 450, 1)$

$= (-1, 0, -350, 0)$

$w_{new} = (-1, 0, -350, 0) \quad b_{new} = 0$

Iteration 3

$x = (4, 2, 150, 1), \quad y = (+1)$

$f(x) = (-1 \times 4) + (0 \times 2) + (-350 \times 150) + (1 \times 0) + 0$

$= -4 + -52500 + 0 \Rightarrow (-52504)$

$y \cdot f(x) = (+1)(-52504) < 0$

Update weight

$yx = (4, 2, 150, 1)$

$w - yx = (-1, 0, -350) - (4, 2, 150, 1)$

$= (-5, -2, -500, -1)$

$w = (-1, 0, -350, 0) - (-5, -2, -500, -1)$

$= (4, 2, 150, 1), \quad b = 1$

$$\text{Margin} = \frac{2}{\|w\|}$$

$$w = (4, 2, 150, 1)$$

$$\|w\| = \sqrt{w_1^2 + w_2^2 + w_3^2 + w_4^2}$$

$$= \sqrt{4^2 + 2^2 + 150^2 + 1^2}$$

$$= \sqrt{16 + 4 + 22500 + 1}$$

$$= 150.07$$

$$\text{Margin} = \frac{2}{150.07} = 0.0133$$